Deep Learning Study Notes

Sample document for testing PDF text extraction in your AI Personal Study Assistant

1. Introduction to Deep Learning

Deep learning is a branch of machine learning that uses artificial neural networks with multiple layers to learn patterns from data. These networks can automatically learn useful representations from examples such as images, audio, and text.

A typical neural network contains an input layer, one or more hidden layers, and an output layer. During training, the network adjusts its weights to reduce the difference between predicted outputs and the correct answers.

Important terms

- **Neuron:** A computational unit that combines inputs, weights, and a bias before applying an activation function.
- **Weight:** A learnable parameter that controls the influence of an input.
- **Bias:** An additional learnable value that shifts a neuron's activation.
- **Activation function:** A function such as ReLU or sigmoid that introduces non-linearity.

2. Convolutional Neural Networks (CNNs)

A Convolutional Neural Network (CNN) is a deep learning architecture commonly used for image-related tasks. CNNs learn visual features such as edges, textures, shapes, and object parts by applying filters to input images.

A CNN commonly contains convolution layers, activation functions, pooling layers, and fully connected layers. The exact architecture depends on the task and model design.

3. Main Layers in a CNN

Convolution layer

A convolution layer applies learnable filters, also called kernels, across an input image or feature map. Each filter responds to patterns in a local region and produces a feature map. Early layers may learn simple patterns such as edges, while deeper layers may learn more complex patterns.

Activation function

The Rectified Linear Unit (ReLU) is a commonly used activation function. It outputs zero for negative inputs and returns the input unchanged for positive inputs. This helps a network learn non-linear relationships.

Pooling layer

Pooling reduces the spatial dimensions of feature maps. Max pooling selects the maximum value from each small region, while average pooling calculates the average value. Pooling can reduce computation and make representations less sensitive to small shifts in the input.

Fully connected layer

A fully connected layer combines features learned by earlier layers to produce predictions. In image classification, the final layer often produces class scores or probabilities.

4. CNN Processing Example

Consider an image classifier that distinguishes cats from dogs. The input image passes through convolution layers that detect visual patterns. Pooling layers reduce feature-map dimensions. Later layers combine the learned features, and the output layer predicts a class such as cat or dog.

5. Training a Neural Network

Training usually involves a forward pass, loss calculation, backpropagation, and parameter updates. During the forward pass, input data moves through the network to produce a prediction. A loss function measures prediction error. Backpropagation calculates gradients, and an optimizer updates the weights.

Common terms

- **Epoch:** One complete pass through the training dataset.
- **Batch:** A subset of training examples processed together.
- **Learning rate:** A value that controls the size of parameter updates.
- **Overfitting:** When a model performs well on training data but poorly on unseen data.

6. Quick Revision Questions

1. What is deep learning?
2. What is the role of a convolution layer in a CNN?
3. How does max pooling work?
4. What is the purpose of an activation function?
5. What is overfitting, and how can it affect model performance?

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